



Non-IID Data Challenges

What's hard

- Client data are skewed and imbalanced; distributions drift over time

Diagnose

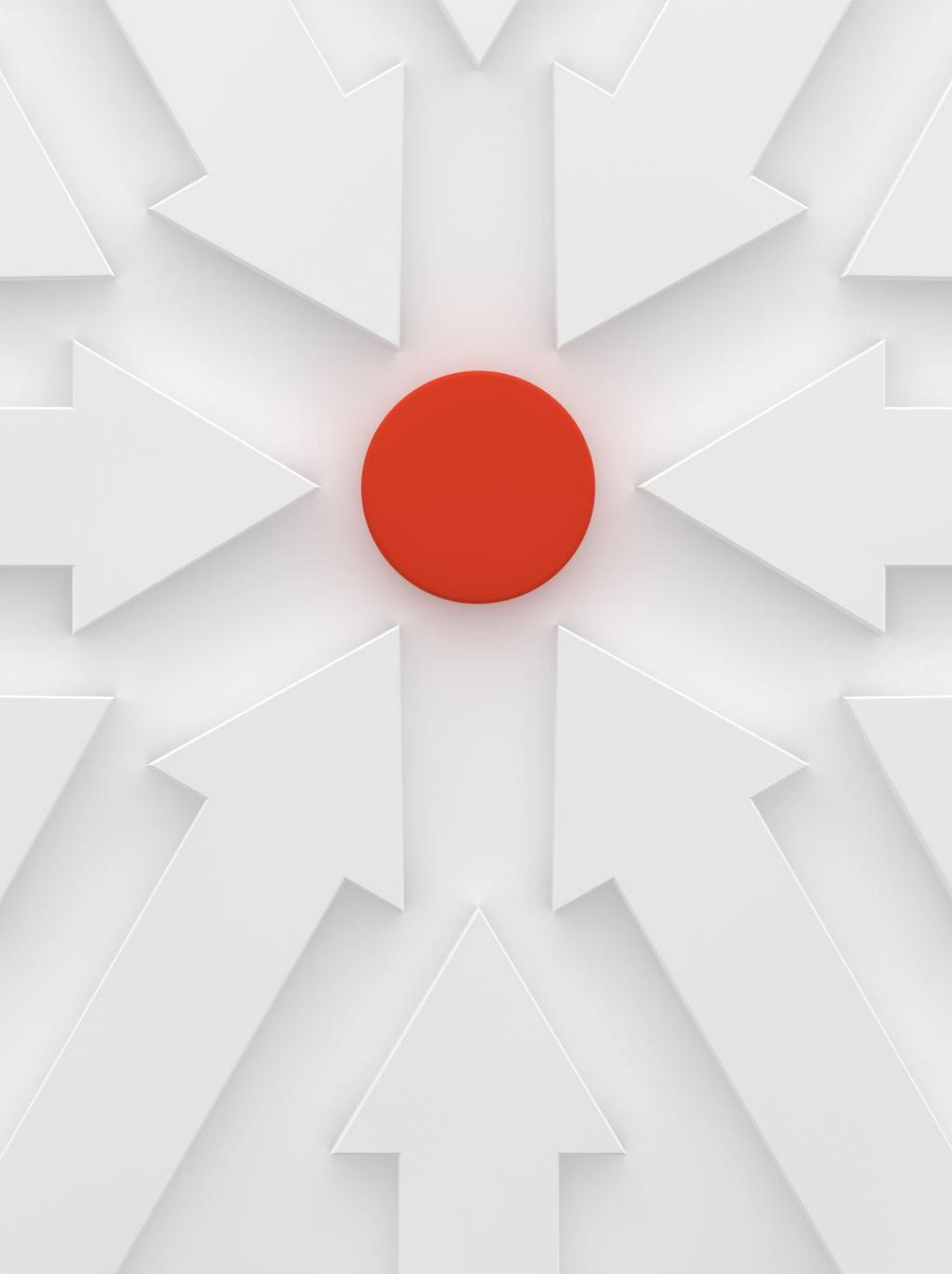
- Compute per-client stats (label/feature histograms, entropy, KS tests)
- Cluster similar clients; track drift across rounds

Adapt aggregation

- Weight by data size, quality, and similarity to global
- Use cohort-wise or personalized aggregation instead of one-size-fits-all

Practical guardrails

- Robustness: clip norms, outlier filtering
- Evaluation: group metrics and worst-case performance



Federated Learning Performance

Convergence factors

- Local steps vs. rounds; data and system heterogeneity; client sampling

Efficiency trade-offs

- More local steps → fewer rounds but risk of drift
- Communication cost vs. accuracy curve

Key metrics

- Rounds to target accuracy; bandwidth per round/total
- Participation rate and straggler impact
- On-device energy/time per round



Differential Privacy Mechanisms

- **Definition**
 - (ϵ, δ) -DP: output barely changes when one record is added/removed
- **Core ingredients**
 - Clip per-client update to bound sensitivity
 - Add calibrated Gaussian noise; track privacy with an accountant
- **In federated learning**
 - Do clipping + noise on device; combine with secure aggregation
 - Monitor spent ϵ, δ ; stop or adjust when budget nears limit
- **Practical tuning**
 - Set clip norm from quantiles; choose noise to hit target ϵ
 - Larger batches and fewer rounds reduce privacy cost

Performance Optimization

Smart client selection

- Score by data utility, device health, connectivity; diversify to avoid bias

Communication scheduling

- Prefer Wi-Fi + charging; predict availability windows; cache and batch updates

Adaptive training

- Per-device configs: batch size, local steps, precision; auto-tune online
- Early exit for slow/overheating devices

Bandwidth/compute efficiency

- Update compression: sparsify, quantize, delta + server-side momentum
- Partial model updates or adapters; server-side aggregation acceleration

Energy awareness

- Train only on charge or high battery; throttle CPU/GPU; budget per session

Metrics to track

- Time/round, completion rate, bytes/round, on-device energy, accuracy/round

